

END-TERM PROJECT-01

Game Programming with C++ (CSE 3545)

Programme: B.Tech., CSE

Semester: 6th

Posted on: 09/04/2026

Submission deadline: Last week of Endsem

Subject Learning Outcomes/ Course Outcomes	*Taxonomy Level	Question Number	Marks
Write, compile and execute simple C++ programs to solve computational problems.	L6		10
Become familiar with basic games using SFML library			
Implement Game Mechanics and Physics.			
Apply advanced object-oriented programming concepts like inheritance, polymorphism, and abstraction			
Develop advanced game features like sprite sheets, vertex arrays, texture management, and layering views.			
Refine the game using file I/O, sound effects, HUD & UI.			

*Bloom's taxonomy levels: Knowledge (L1), Comprehension (L2), Application (L3), Analysis (L4), Evaluation (L5), Creation (L6)

Car Dodging Game

Design and develop a 2D Car Dodging Game using C++ and SFML (Simple and Fast Multimedia Library). The game simulates a car driving along a vertical road where the player must avoid collisions with incoming vehicles. The player controls a car (white block) that can move left and right across lanes, while enemy cars (red blocks) continuously spawn from the top and move downward. The objective is to survive as long as possible and achieve the highest score.

Basic Set of Features and Game Mechanics

1. Apply Object-Oriented Programming concepts: Classes like Car, PlayerCar, EnemyCar using inheritance.
2. Player Car Control
 - Move left and right using keyboard input (e.g., arrow keys or A/D).
 - Restrict movement within road boundaries.
3. Enemy Cars Control
 - Spawn randomly in different lanes.

- Move downward at increasing speed.
- Remove cars once they go off-screen.

4. Collision Detection

- Detect collision between player car and enemy cars.
- End game upon collision.

5. Score System

- Increase score based on survival time or number of dodged cars.
- Display score on screen.

6. Speed Mechanism

- Gradually increase game speed over time.
- Display current speed level and real speed.

7. Game Window

- Fullscreen or fixed resolution window.
- Render road (lane markings), player car, and enemy cars.

8. Difficulty level

- Easy, Medium, Hard with respect to i) Increasing traffic density over time ii) Randomized enemy car speeds

9. Audio & Visuals

- Background music and sound effects (collision, movement)
- Improved graphics using textures

10. Game Functionalities

- Pause and resume functionality
- Restart option after game over

Deliverables

1. Source code
2. Executable file
3. Short report including:

- (a) Game design
- (b) Features implemented
- (c) Challenges faced

Sample Output of the Project

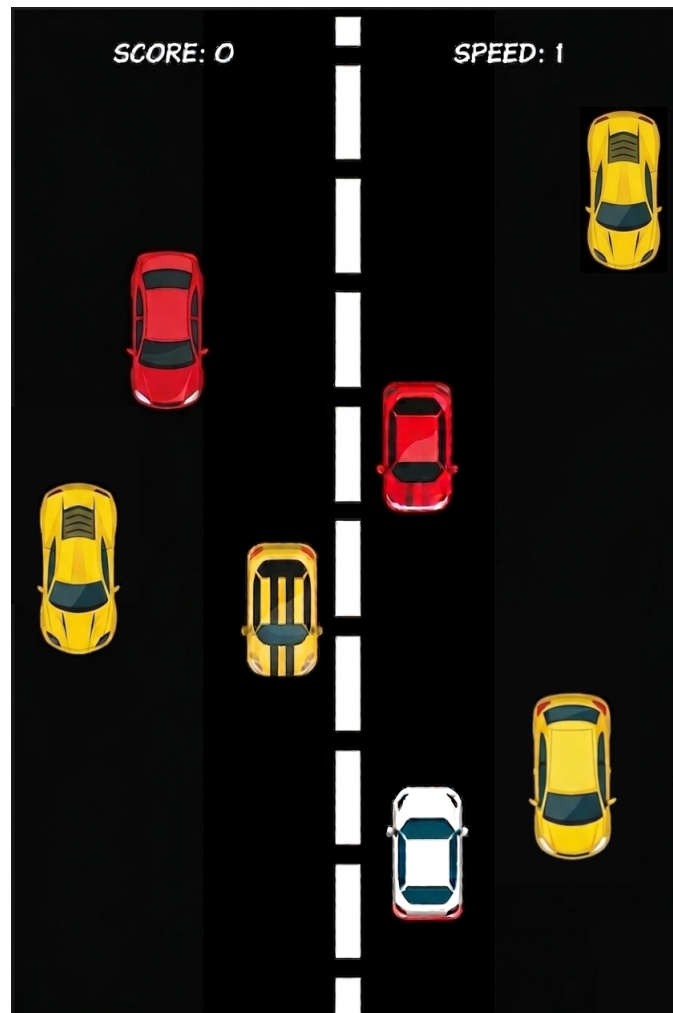


Figure 1: Playable game Screen, Player Car: White, Enemy Cars: Any other color

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific outcomes (PSOs)

COs	POs											PSOs	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO1	3	2			2						2	3	2
CO2	2		2		3							3	2
CO3	3	3	3	2								3	2
CO4	3	2	3					2			2	3	2
CO5	3		3	2	3							3	3
CO6	2		2		3				2		3	3	3